

Section 11-5:

Confidence Interval:

A confidence interval for slope β_1 can be computed by using the following formula.

$$b_1 - t_{\alpha/2} \frac{s}{\sqrt{SS_{xx}}} < \beta_1 < b_1 + t_{\alpha/2} \frac{s}{\sqrt{SS_{xx}}}$$

$b_1 \rightarrow$ Estimated slope

$s \rightarrow$ standard error of estimate

$SS_{xx} \rightarrow$ sum of squared deviations

$t_{\alpha/2} \rightarrow$ critical value of t dist for confidence interval.

$$\Rightarrow s^2 = \frac{SS_{yy} - b_1 SS_{xy}}{n-2} \text{ or } SSE = \frac{\sum (y - \hat{y})^2}{n-2}$$

$$\Rightarrow SS_{xx} = \frac{\sum x^2 - (\sum x)^2}{n}$$

$$\Rightarrow SS_{yy} = \frac{\sum y^2 - (\sum y)^2}{n}$$

$$\Rightarrow SS_{xy} = \frac{\sum xy - \sum x \sum y}{n}$$

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Hypothesis Testing for slope.

=> Hypothesis:

$$H_0 : \beta_1 = \beta_{10}$$

$$H_1 : \beta_1 \neq \beta_{10}$$

$$H_1 : \beta_1 < \beta_{10}$$

$$H_1 : \beta_1 > \beta_{10}$$

=> Test Statistics:

$$t_0 = \frac{b_1 - \beta_{10}}{S / \sqrt{SS_{xx}}}$$

=> Critical Region:

$$\begin{cases} t \geq t_{\alpha/2}(v) \\ t \leq -t_{\alpha/2}(v) \end{cases}$$

$$t > t_{\alpha}(v)$$

$$t < -t_{\alpha}(v)$$

$$v = n - 2.$$

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Quality of Fit:

Coefficient of Determination (R^2)

$\Rightarrow R^2$, the coefficient of determination, is a measure of how well the regression model fits the data. It is defined as,

$$R^2 = \cancel{1 - \frac{SSE}{SST}} \quad 1 - \frac{SSE}{SST}$$

$$SSE = \sum_{i=1}^n (y_i - \hat{y})^2$$

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$SSR = SST - SSE$$

High R^2 means good fit, while a low value suggests that the model does not explain much of variability in the data.

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Confidence Interval for β_0

$$b_0 - t_{\alpha/2} S \sqrt{\frac{\sum x_i^2}{nSS_{xx}}} < \beta_0 < b_0 + t_{\alpha/2} S \sqrt{\frac{\sum x_i^2}{nSS_{xx}}}$$

$$n = v - 2.$$

Hypothesis Testing.

Hypothesis:

$$H_0 : \beta_0 = \beta_{00}$$

$$H_1 : \beta_0 \neq \beta_{00}$$

$$H_1 : \beta_0 < \beta_{00}$$

$$H_1 : \beta_0 > \beta_{00}$$

Test Statistics :

$$t = \frac{b_0 - \beta_{00}}{S \sqrt{\frac{\sum x_i^2}{nSS_{xx}}}}$$

Critical Region

$$\left\{ \begin{array}{l} t \leq -t_{\alpha/2} \\ t \geq t_{\alpha/2} \end{array} \right\}$$

$$t < -t_{\alpha}$$

$$t > t_{\alpha}$$

Question 1:

Find a 95% confidence interval for β_1 in the regression line $\mu_{Y|x} = \beta_0 + \beta_1 x$, based on the pollution data of Table 11.1.

Table 11.1: Measures of Reduction in Solids and Oxygen Demand

Solids Reduction, x (%)	Oxygen Demand Reduction, y (%)	Solids Reduction, x (%)	Oxygen Demand Reduction, y (%)
3	5	36	34
7	11	37	36
11	21	38	38
15	16	39	37
18	16	39	36
27	28	39	45
29	27	40	39
30	25	41	41
30	35	42	40
31	30	42	44
31	40	43	37
32	32	44	44
33	34	45	46
33	32	46	46
34	34	47	49
36	37	50	51
36	38		

Confidence interval
 $\alpha = 0.05$. $v = n - 2 = 31$

$$b_1 - t_{\alpha/2} S \leq \beta_1 \leq b_1 + t_{\alpha/2} S \quad (1)$$

$$\Rightarrow S^2 = \frac{SS_{yy} - b_1 SS_{xy}}{n - 2}$$

$$SS_{yy} = \sum y^2 - \frac{(\sum y)^2}{n} = 3713.88$$

$$SS_{xy} = (3752.09)$$

$$SS_{xx} = 4152.18$$

$$S^2 = \frac{3713.88 - (0.903643)(3752.09)}{31}$$

$$S^2 = 10.4299$$

$$S = 3.2295$$

$$b_1 = \frac{SS_{xy}}{SS_{xx}} = \frac{(3752.09)}{4152.18}$$

$$= 0.9036$$

Put in eq ①

$$t_{0.05/2}(31) = 2.045.$$

$$0.903643 - \frac{(2.045)(3.2295)}{\sqrt{4152.18}} < \beta < 0.903643 + \frac{(2.045)(3.2295)}{\sqrt{4152.18}}$$

$$0.8012 < \beta_1 < 1.0061$$

Question 2:

Using the estimated value $b_1 = 0.903643$ of Question 1, test the hypothesis that $\beta_1 = 1.0$ against the alternative that $\beta_1 < 1.0$. With 95% confidence interval

Hypothesis:

$$H_0: \beta_1 = 1.0$$

$$H_1: \beta_1 < 1.0$$

Test statistics:

$$t = \frac{b_1 - \beta_{10}}{s / \sqrt{SS_{xx}}}$$

$$t = \frac{0.903643 - 1.0}{3.2295 / \sqrt{4152.18}}$$

$$= -1.92$$

Critical Region:

$$v = 31 \quad \alpha = 0.05.$$

$$t_{0.05}(31) = 1.6905.$$

$$t < -t_{\alpha} \quad -1.92 < -1.69$$

$$1.69 < 1.92$$

Reject H_0

We can say that the estimated slope is lower than 1

Question 3:

Find a 95% confidence interval for β_0 in the regression line $\mu_{Y|x} = \beta_0 + \beta_1 x$, based on the data of question 1

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Confidence interval

$$b_0 - t_{\alpha/2} s \sqrt{\frac{\sum x_i^2}{n \sum s_{xx}}} < \beta_0 < b_0 + t_{\alpha/2} s \sqrt{\frac{\sum x_i^2}{n \sum s_{xx}}}$$

from previous question
 $\sum s_{xx} = 4152.18$
 $s = 3.2295$
 $\sum x_i^2 = 41,086$
 $b_0 = 3.829633$
put values
 $t_{0.05/2}(31) = 2.045$

$$3.829633 - (2.045)(3.2295) \sqrt{\frac{41,086}{(33)(4152.18)}} < \beta_0 < \dots$$

$= 0.2132 < \beta_0 < 7.4461$

Question 4: Using the estimated value $b_0 = 3.829633$ of question 1, test the hypothesis that $\beta_0 = 0$ at the 0.05 level of significance against the alternative that $\beta_0 \neq 0$.

Hypothesis:

$$H_0 : \beta_0 = 0$$

$$H_1 : \beta_0 \neq 0$$

Test Stats:

$$t = \frac{b_0 - \beta_{00}}{\frac{s \sqrt{\sum x_i^2}}{\sqrt{n(SS_{xx})}}}$$

From prev question

$$\begin{aligned} t &= \frac{3.829633 - 0}{3.2295 \sqrt{\frac{41,086}{(33)(4152.18)}}} \\ &= 2.17. \end{aligned}$$

Critical Region:

$$t_{0.025}(31) = 2.03.$$

$$2.17 \leq -2.03$$

$$2.17 \geq 2.03 \quad \text{True}$$

H_0 Reject.